ISG-NIS-Helper — Usage

Day-to-day running: managing who has access, reading what it recorded, and changing settings.

1. What a user sees

They open NIS-Elements the way they always have — the desktop icon, the Start menu, or the taskbar. Instead of NIS opening straight away, a window asks for their name and PIN, with a line saying the instrument logs all usage and training is required. They tick a confirmation and sign in, and only then does NIS-Elements start. A small badge in the corner shows who is signed in and for how long. (Cancelling the sign-in simply does not open NIS.)

The session ends when NIS-Elements closes. There is no timeout, no idle check and no automatic logout — if NIS is open, the session is open, for ten minutes or four days.

Nothing on screen carries the project name. Everything is titled with `[instrument] display_name`, default **Microscope Access Log**.

That is not a security measure and should not be described as one. Task Manager shows the executables, Task Scheduler shows the tasks, and everyone on these machines is an administrator. What it buys is salience: a product name invites someone to search for it and start poking; a plain instrument-ish window gets ignored.

2. The approved user list

`users.csv` in the ISG-NIS-Helper folder is the training gate. **One list per machine, kept with the program.** There is no network dependency, so there is no stale-cache failure mode: either the file is there and readable, or it is not and ISG-NIS-Helper says so on the sign-in window.

The installer creates an empty one from `users_template.csv` if there isn't one. Preflight fails a list with no active users.

```
name,email,pin,active,Lab,notes
Andrew Bazley,Andrew.Bazley@nyulangone.org,7161,yes,Holt,instrument manager
```

ISG-NIS-Helper reads `name`, `email`, `pin`, `active` and `notes`. Any other column you add is ignored and left untouched — ISG-NIS-Helper never writes to this file.

To add somebody: add a row with a PIN nobody else on that machine has, `active` set to `yes`, and the training date in `notes`. Picked up at the next sign-in; nothing needs restarting.

To remove somebody: set `active` to `no`. **Do not delete the row** — their name needs to stay resolvable when you read a log from eight months ago.

Four machines means four lists. That is the trade for having no network dependency, and with a dozen users it is a small one. Keep a master copy wherever suits you and copy it out when it changes; treat that as a deliberate act rather than something that happens silently behind you. Run preflight afterwards and it will tell you if two people ended up sharing a PIN, which destroys attribution.

With the recommended `C:\ISG-NIS-Helper` location, you can edit `users.csv` in Notepad normally. (Only if you installed under `C:\Program Files` does Notepad need to be run as administrator to save.)

PINs. You assign the first one. On first sign-in each person is offered the chance to replace it with one only they know, and may skip. Overrides are stored as one small file per person under `C:\ProgramData\ISG-NIS-Helper\pins\` — not in `users.csv`, which stays entirely yours. To reset somebody, delete their file from there.

PINs are plain text on purpose. This is a deterrent, not a credential system, and a PIN nobody can reset is worse than one somebody can read. Tell people not to reuse a PIN that matters.

The master PIN is per-instrument, generated by the pre-scan, stored in that machine's `config.ini`. Circulate it to department managers only. Using it requires typing a name and flags the session `master_pin_used` in the log. It exists so a manager is never locked out, not as a routine way in.

3. What a session produces

Three files per session, in `C:\ProgramData\ISG-NIS-Helper\logs\<PC>\<year>\<month>\` and mirrored to the share. Nothing is appended to or overwritten.

```
2026-09-21_1432_Bazley-Andrew_074m_a1b2c3d4.csv           the session row
2026-09-21_1432_Bazley-Andrew_074m_a1b2c3d4_events.jsonl  forensics
2026-09-21_1432_Bazley-Andrew_074m_a1b2c3d4_files.csv     files changed
```

The name carries date, start time, surname-first user, duration in minutes, and session ID.

The session row: name and email, start and end in local time and UTC, duration, how it ended, whether NIS was started through ISG-NIS-Helper or directly, whether the master PIN was used, counts of files new/modified/deleted, bytes added, and whether the user list came from the share or the local cache.

The events file: failed PINs, unauthorized launches and whether they were closed, the watcher starting and stopping, the share going away, a snapshot failing, somebody clicking "request training".

The file list: every file created, modified or deleted under the data roots during that session. Its purpose is to tell you which ND2 files and which NIS logs to open when something has gone wrong, and roughly when.

An optional rolling `index.csv` per computer collects every session row in one skimmable file. Convenience only; safe to delete.

If you set `[paths] network_root`, finished session files are also **copied** to that folder so several scopes pool their records. It is a mirror and nothing more: ISG-NIS-Helper never reads from it and never depends on it, and if it is unreachable the copies queue locally and go across later. Leaving it blank is a complete, working configuration.

4. Reading the logs

Every session on a PC, newest first:

```
Import-Csv C:\ProgramData\ISG-NIS-Helper\logs\<PC>\index.csv |
  Sort-Object start_local -Descending |
  Format-Table start_local, user_name, duration_min, files_new, end_reason
```

Every unauthorized launch:

```
Get-ChildItem C:\ProgramData\ISG-NIS-Helper\logs\<PC> -Recurse -Filter *_events.jsonl |
  Get-Content | ConvertFrom-Json |
  Where-Object kind -like 'unauthorized*'
```

Who was on the scope when a particular file was written:

```
Get-ChildItem C:\ProgramData\ISG-NIS-Helper\logs\<PC> -Recurse -Filter *_files.csv |
  ForEach-Object { $f = $_; Import-Csv $_ |
    Where-Object path -like '*the-file-name*' |
    Select-Object @{n='session';e={$f.Name}}, change, path }
```

5. Changing settings

Edit `config.ini` in the program folder, then restart the agent so it re-reads it:

```
schtasks /End /TN ISG-NIS-Helper-Agent
schtasks /Run /TN ISG-NIS-Helper-Agent
```

With `C:\ISG-NIS-Helper` this saves normally; only under `C:\Program Files` must Notepad be run as administrator or the save silently fails.

Changing `network_root` — for instance when you move from a local folder to the lab share — is just this one line plus the restart. Nothing needs reinstalling.

Run `ISG-NIS-Helper-Preflight.ps1` afterwards. It checks that no placeholders remain, the NIS executable exists, **the executable ISG-NIS-Helper launches is in the list of process names it watches** (get that wrong and sessions never end), every data root exists, the share is reachable and writable, `users.csv` parses, no two people share a PIN, the log store is writable, both tasks are registered, and the clock is synchronised.

6. What it cannot do

It cannot tell who is at the keyboard. A trained user who signs in and hands over the mouse defeats it completely. That is not a bug to be fixed; it is the reason the log exists. The policy it enables is: *whoever is signed in is responsible for the instrument for the duration of their session, including for anyone they allow to operate it.*

Everyone on these machines is an administrator, so the watcher can be ended from Task Manager. It restarts within a minute and the gap is visible in the events file, which makes circumvention deliberate and legible rather than impossible. Reducing the daily account to a standard user would prevent more damage than the sign-in screen does, and is worth testing on one machine.

Clocks on standalone PCs drift. Both local time and UTC are recorded so drift is detectable, but see the Installation guide.

The file diff sees everything that changed under the data roots, including files somebody moved in Explorer with NIS closed. It is an index of what changed, not a claim about what NIS did.

A very large data root makes snapshots slow. They run in the background and never block anyone, but if a root holds millions of files, narrow `data_roots` or set `include_extensions`.

7. Folder layout

On the microscope PC:

```
C:\ISG-NIS-Helper\                 everything about the program
    config.ini                   this machine's settings
    users.csv                    this machine's approved users
    docs\                        these guides
    install-logs\                build, install and diagnostic records
C:\ProgramData\ISG-NIS-Helper\       everything the program writes
    session.json                 the open session, if any
    pins\                        self-chosen PIN overrides
    agent.json                   the sign-in program's beacon
    health.json                  failsafe state, rewritten every 10 minutes
    SAFEMODE.json                present only when enforcement is disabled
    logs\<PC>\index.csv          rolling index, optional
    logs\<PC>\<year>\<month>\    the per-session files
    logs\<PC>\events_rolling\    one events file per day, always written
```

If `network_root` is set, that folder receives copies of finished session files under `logs\<PC>\<year>\<month>\` and nothing else.